

Erdős Problem #161

EP-161

L1 - TRACTABLE

PARTIALLY SOLVED

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $\alpha \in [0, 1/2)$ and $n, t \geq 1$. Let $F^{(t)}(n, \alpha)$ be the smallest m such that we can 2-colour the edges of the complete t -uniform hypergraph on n vertices such that if $X \subseteq [n]$ with $|X| \geq m$ then there are at least $\alpha \binom{|X|}{t}$ many t -subsets of X of each colour.

For fixed n, t as we change α from 0 to $1/2$ does $F^{(t)}(n, \alpha)$ increase continuously or are there jumps?
Only one jump?

Source and attribution. Statement reproduced from UnsolvedMath, whose official dataset is released under the CC BY 4.0 license.

Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>